

Title

Sensorimotor circuit connectivity as a candidate biomarker for responsiveness to selective serotonin reuptake inhibitors in obsessive–compulsive disorder

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Abstract

Predicting selective serotonin reuptake inhibitor (SSRI) response in obsessive–compulsive disorder (OCD) remains a clinical challenge. As the sensorimotor circuit is functionally altered in OCD and is susceptible to serotonergic modulation, we hypothesized that its baseline functional connectivity (FC) could serve as a predictive biomarker for treatment response. We enrolled 54 drug-naïve patients with OCD and 39 matched healthy controls (HC), who then underwent resting-state fMRI. Patients received sertraline for 12 weeks and were classified as responders (rOCD, $n = 33$) or non-responders (nOCD, $n = 21$) based on reductions in Yale-Brown Obsessive Compulsive Scale scores. Seed-based FC analysis of the sensorimotor circuit was conducted across the three groups. We found that OCD patients exhibited abnormal FC primarily within the sensorimotor circuit and in its connections with the cerebellum. Notably, the rOCD group showed increased FC values within the sensorimotor circuit relative to HC, whereas the nOCD group showed reduced FC values. The cerebellar regions that exhibited altered FC were involved in both sensorimotor processing and non-sensorimotor functions, such as salience detection and executive control. All analyzed FC values were significantly associated with SSRI treatment response (FDR-adjusted $p < 0.05$), with R^2 values between 0.144 and 0.479, indicating moderate to high explanatory power. Taken together, baseline sensorimotor-circuit FC differentiates responders from non-responders and holds promise as a neuroimaging biomarker to guide individualized pharmacological treatment in OCD.

Introduction

Obsessive-compulsive disorder (OCD) is a chronic and disabling psychiatric disorder, characterized by distress-inducing intrusive thoughts (obsessions) and repetitive or ritualized behaviors (compulsions)[1]. Epidemiological studies indicate a global prevalence of approximately 2%-3%[2], with a lifetime prevalence in China reaching 2.4%[3]. Affected individuals experience significant functional impairment[4] and impose substantial socioeconomic burdens[5]. However, approximately half of patients respond to first-line pharmacological strategies[6]—selective serotonin reuptake inhibitors (SSRIs)—with complete symptomatic remission observed in even fewer cases[7]. In addition, reliance on post hoc identification through sequential medication trials delays effective intervention and exacerbates disability burden. Although some clinical features—such as poor insight, religious or hoarding symptoms, and high comorbidity—have been linked to poor response[8], they offer limited predictive power and lack mechanistic specificity, highlighting the need for clinically applicable biomarkers capable of prospectively distinguishing SSRI responders from non-responders.

Pharmacological strategies for OCD are limited by an incomplete understanding of its pathological mechanisms. The most extensively validated dysfunctional network in OCD is the cortico-striato-thalamo-cortical (CSTC) loop, which governs processes such as habitual behavior, emotional regulation, and the integration of sensory and motor information[9,10]. The CSTC loop is composed of parallel yet partially segregated circuits, including the sensorimotor circuit that primarily connects the premotor cortex, putamen, and thalamus[11,12]. Although historically overlooked, the sensorimotor circuit was recently demonstrated by a mega-analysis from the Enhancing Neuro-Imaging and Genetics through Meta-Analysis (ENIGMA) OCD consortium to exhibit widespread hypoconnectivity across 28 cohorts[13], challenging previous emphasis on other dysfunctional brain networks[14,15]. Notably, dysfunction of the sensorimotor circuit is closely associated with several core clinical features of OCD, such as habitual behaviors, automatic responses and sensory phenomena (SP)[12]. In addition, evidence from neuropsychological research suggests that impaired response inhibition—a potential endophenotypic marker for OCD—engages the cortical and subcortical regions within the sensorimotor circuit[16-19], highlighting a more central role for sensorimotor circuit in the neurobiology of OCD.

Moreover, the sensorimotor circuit is structurally connected to the serotonergic system and may serve as a functional substrate for SSRI modulation. The raphe nuclei (RNi)—the primary source of 5-hydroxytryptamine (5-HT)—project densely to key

nodes of the sensorimotor circuit, including thalamus and sensory cortices[20,21]. In addition, preclinical studies revealed that agonists of the 5-HT_{1B} receptor—a presynaptic autoreceptor on serotonergic neurons—could lead to increased perseverative behaviors and impairments in sensorimotor gating in rodents, which could be attenuated by chronic SSRI treatment[22,23]. A similar correlation between 5-HT_{1B} receptor and dysfunctional sensorimotor gating was also observed in OCD patients using positron emission tomography (PET)[24]. Moreover, converging evidence further indicates that SSRIs can broaden network-level plasticity in circuits encompassing the sensorimotor circuit, enabling long-term potentiation (LTP)-like plasticity and longer-lasting responsiveness to neuromodulatory inputs[25]. Taken together, the evidence positions the sensorimotor circuit as a mechanistically plausible substrate through which serotonergic modulation may influence treatment outcomes.

Recent studies have used indices derived from the resting-state functional magnetic resonance imaging (rs-fMRI), such as functional connectivity (FC), to predict SSRI response in OCD, with encouraging findings reported across the studies[26,27]. However, the selected seed regions either related to serotonergic pharmacology, such as RNi[26], or implicated in the putative mechanisms of OCD alone, such as the cerebellum[27], limiting disorder specificity and mechanistic interpretability. To address this gap, we focus on the sensorimotor circuit as a prespecified seed region, given its position at the intersection of core OCD pathophysiology and the pharmacological effects of SSRIs. Additionally, without imposing seeds, whole-brain rs-fMRI studies have converged on the sensorimotor circuit as a prominent predictor of SSRI response[28,29]. For example, voxel-mirrored homotopic connectivity (VMHC) in the postcentral gyrus was found to be associated with SSRI efficacy in medication-free OCD patients[28]. Together, this evidence supports the sensorimotor circuit as a candidate that can offer disorder-relevant, mechanistically interpretable predictive value.

To address the limitations, we employed a seed-based FC approach to examine alterations of the sensorimotor circuit in drug-naïve OCD patients. We aimed to identify baseline differences in FC between SSRI responders (rOCD) and non-responders (nOCD), and compare both groups to healthy controls (HC). Furthermore, we evaluated whether baseline FC values could serve as predictors of SSRI treatment response, using binary logistic regression. We hypothesized that sensorimotor connectivity at baseline would differentiate rOCD from nOCD and serve as a candidate biomarker for SSRI response.

Materials and Methods

Participants and research procedure

A total of 59 drug-naïve OCD outpatients were recruited from Shanghai Mental Health Center (SMHC) in China from September 2020 to August 2023. Inclusion criteria for patients were: (1) aged 18-65 years; (2) diagnosed by psychiatrists following the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) criteria[30]; (3) Yale-Brown Obsessive Compulsive Scale (Y-BOCS) score ≥ 20 [31]; (4) no prior medication or any form of psychotherapy for OCD in the past 6 months. Forty-six age-, sex- and education level-matched HC were recruited through local advertisements, with inclusion criteria as follows: (1) aged 18-65 years; (2) no personal or first-degree family history of mental illness.

Exclusion criteria for both patients and HC were: (1) meeting DSM-5 diagnostic criteria for Schizophrenia Spectrum and Other Psychotic Disorders, or Bipolar and Related Disorders; (2) moderate or higher risk of suicide; (3) substance use in the past year severe enough to potentially impact treatment adherence; (4) severe depression with Beck Depression Inventory (BDI) score ≥ 29 [32]; (5) comorbid psychiatric or medical disorders likely to impact medication adherence or efficacy (e.g., borderline personality disorder, CNS disorders); (6) pregnancy or lactation.

The data analyzed in this study were obtained from an existing clinical cohort (NCT04539951), with detailed methodology described in the original protocol[33]. All participants underwent rs-fMRI scanning and provided demographic data at baseline. OCD patients subsequently received sertraline, initially at 50 mg/d, with weekly 50 mg/d increments up to the maximum recommended dosage (200 mg/d) or the maximum tolerated dosage (<200 mg/d). OCD symptom severities were assessed using the Y-BOCS at baseline and after 12 weeks of treatment. Treatment response (rOCD) was defined as a $\geq 35\%$ reduction from baseline in Y-BOCS scores following treatment, whereas non-response (nOCD) was defined as a $<35\%$ reduction from baseline[34]. The study was approved by the Research Ethics Committee at Shanghai Mental Health Center, Shanghai Jiao Tong University School of Medicine. Participants provided written informed consent before enrollment.

Image acquisition

MRI data were acquired using a Siemens 3T Verio scanner at SMHC. Participants were instructed to relax, remain awake, and refrain from active thinking. Foam pads and earplugs were provided to reduce head motions and scanner noise. High-resolution anatomical images were acquired using a three-dimensional magnetization-prepared rapid acquisition gradient echo (3D MPRAGE) sequence with the following parameters: repetition time (TR) = 2300 ms, echo time (TE) = 3.5 ms,

flip angle = 9° , matrix size = 64×64 , field of view (FOV) = $256 \text{ mm} \times 256 \text{ mm}$, slice thickness = 1 mm, number of slices = 192, acquisition voxel size = $1 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$.

Resting-state BOLD-fMRI data were collected using a standard gradient echo planar imaging sequence with the following parameters: TR = 2000 ms, TE = 30 ms, flip angle = 77° , matrix size = 64×64 , FOV = $220 \text{ mm} \times 220 \text{ mm}$, voxel size = $3 \text{ mm} \times 3 \text{ mm} \times 3 \text{ mm}$. 50 axial slices with a slice thickness of 3 mm, and no slice gap. A total of 240 volumes were obtained over 480 seconds.

Data preprocessing

Preprocessing and quality assurance were performed by CONN Toolbox version 22.v[35], based on SPM version 12 (SPM12) software[36]. Structural images were visually inspected using MRIcron software, and subjects with excessive deformation, artifacts, or incomplete scans were excluded. Images were preprocessed as follows: Functional images were first realigned and unwarped to correct for head motion and susceptibility distortion interactions, with all volumes coregistered to the first scan. Outlier volumes were detected using the Artifact Detection Tool (ART), defined as scans with framewise displacement $> 0.9 \text{ mm}$ or global BOLD signal changes > 5 standard deviations. Functional and anatomical images were then segmented into gray matter, white matter, and CSF, normalized into standard MNI space using SPM's unified segmentation and normalization approach, and resampled to 2 mm isotropic resolution. Spatial smoothing was applied using a 6 mm full-width-at-half-maximum (FWHM) Gaussian kernel. Subsequently, denoising was performed by regressing out confounding effects, including five principal components each from white matter and CSF (CompCor), 12 motion parameters (6 realignment parameters and their first-order derivatives), identified outlier scans, session effects, and linear trends. Finally, temporal bandpass filtering (0.017–0.1 Hz) was applied to reduce low-frequency drift and high-frequency noise.

In total, 4 patients in the nOCD group, 1 patient in the rOCD group and 7 HC were excluded due to excessive head motion, leaving a final sample of 21 patients in the nOCD group, 33 patients in the rOCD group and 39 HC.

Functional connectivity analysis

FC analyses were conducted using CONN Toolbox. A total of 28 region of interests (ROI) were delineated using a combination of atlases: 24 ROIs matched to sensorimotor cortices from the 200-parcel Kong 2022 Atlas[37], and 4 bilateral subcortical ROIs (left/right thalamus and putamen) from the Harvard-Oxford

Atlas[38], resulting in 14 ROIs per hemisphere. Seed-based whole-brain FC maps were generated by computing Pearson correlation coefficients between the mean time series of each seed ROI and the time series of all other brain voxels. These correlation values were then transformed to Fisher's z-scores for subsequent group-level analyses. A one-way analysis of variance (ANOVA) was performed for each voxel to examine differences in FC among the three groups (nOCD, rOCD, HC) using CONN Toolbox in the general linear model, with the sex, age and years of education as covariates. To determine statistically significant clusters of altered connectivity, we applied a two-step thresholding approach: a voxel-level threshold of $p < 0.001$, followed by a cluster-level family-wise error (FWE) correction at $p < 0.05$ (two-tailed).

Statistical analyses

Chi-square tests, ANOVA tests and independent samples t-tests were used to analyze the demographic and clinical characteristics of the participants using the SPSS software (version 27.0, IBM Corporation, Somers, NY, USA) with a significant threshold of $p < 0.05$ (two-tailed).

Imaging data analyses were conducted using CONN Toolbox. FC values of significant clusters were extracted for each group. Post hoc pairwise comparisons were performed in SPSS with Bonferroni correction, applying a significance threshold of $p < 0.016$ (i.e., $0.05/3$).

Pearson correlation analyses were conducted to assess the associations between differential FC values and clinical features, including both the baseline Y-BOCS scores and the Y-BOCS reduction scores (Δ Y-BOCS), across and within nOCD and rOCD groups. Statistical significance was defined as $p < 0.05$ (two-tailed).

Univariate binary logistic regression analyses were used to identify potential predictors of SSRI treatment response. Each predictor, including clinical features and abnormal FC, was analyzed separately with treatment response as the binary outcome (response vs. non-response). Nagelkerke R^2 , odds ratios (ORs), 95% confidence intervals (CIs), area under the curve (AUC) values and p -values were calculated for each model. False discovery rate (FDR) correction was applied to adjust for multiple comparisons.

Results

Demographic and clinical characteristics

Demographic and clinical characteristics are summarized in Table 1. No significant differences were observed among nOCD, rOCD and HC groups in terms

of sex, age or years of education. Additionally, nOCD and rOCD groups did not differ significantly in illness duration or baseline Y-BOCS scores, while the post-treatment Y-BOCS scores were significantly lower in the rOCD group compared to the nOCD group.

Table 1. Demographic and clinical characteristics of the study participants.

Variables (mean $\pm$ SD)	nOCD(n=21)	rOCD(n=33)	HC(n=39)	$t/F/\chi^2$	p -Value
Sex (male/female)	15/6	19/14	19/20	2.879	0.237 ^a
Age (years)	2	2	3	1.778	0.175 ^b
Education (years)	1	1	1	0.082	0.922 ^b
Illness Duration (years)	1	8	NA	0.916	0.364 ^c
Y-BOCS score					
Before Medication	2	2	NA	1.545	0.128 ^c
After Medication	2	1	NA	9.306	<0.001 ^c

OCD, obsessive-compulsive disorder; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control; Y-BOCS, the Yale-Brown Obsessive Compulsive Scale; NA, not applicable.

^a p -value from the chi-square test; ^b p -value from one-way analysis of variance tests; ^c p -value from independent samples t -tests (nOCD vs. rOCD).

Seed-based functional connectivity analyses

First, we compared FC values between the entire OCD group (nOCD and rOCD) and HC (results in the Supplementary Table 2). Then we conducted a one-way ANOVA of seed-based FC among all three groups (FWE-corrected, $p < 0.05$). Significant between-group differences in FC were then subjected to post hoc analysis, using Bonferroni correction ($p < 0.05$, three comparisons). Table 2 summarizes post hoc differences in FC between nOCD and rOCD groups.

ANOVA results are shown in Figure 1. To avoid repetition, all abbreviations for seed regions and their corresponding brain areas are listed in Supplementary Table 4. In the left hemisphere, the altered FC included Thal_l-FG_r, Thal_l-PoG_l, SomMotA_1_l-VIIb_l, SomMotA_1_l-VI_l, SomMotA_1_l-CC, SomMotA_2_l-STN, SomMotA_2_l-VI_l, SomMotA_2_l-Vermis, SomMotA_2_l-Pu, SomMotA_4_l-VPL_r and SomMotA_4_l-Caud_r.

In the right hemisphere, the altered FC included Thal_r-Crus1_r, Thal_r-PoG_l, SomMotB_4_r-MTG_r and SomMotB_4_r-MOG_l. No significant difference was

found when the right putamen was used as the seed.

Table 2. Brain regions demonstrating different FC values between nOCD and rOCD groups.

Seed region	Brain region	MNI peak			Cluster size	<i>F</i> -value (peak)	Post hoc analyses
		X	Y	Z			
Thalamus_l	Fusiform Gyrus_r	26	-80	-04	129	8.511	nOCD < rOCD, rOCD > HC
	Postcentral Gyrus_r	-44	-28	68	122	9.600	nOCD < rOCD, rOCD > HC
Putamen_l	Extra-nuclear White Matter_r	10	-34	04	86	9.153	nOCD > rOCD, rOCD < HC
SomMotA_1_l	Cerebellar Lobule VIIb_l	-06	-72	-36	439	10.192	nOCD < rOCD, rOCD > HC
	Cerebellar Lobule VI_l	-26	-58	-28	141	8.421	nOCD < rOCD, nOCD < HC
	Corpus Callosum_l	-22	-48	06	122	10.736	nOCD > rOCD, rOCD < HC
SomMotA_2_l	Subthalamic Nucleus_r	10	-14	-08	184	9.536	nOCD < rOCD, rOCD > HC
	Cerebellar Lobule VI_l	-26	-58	-28	167	10.044	nOCD < rOCD, nOCD < HC
	Cerebellar Vermis VII_r	06	-64	-26	85	8.103	nOCD < rOCD
	Pulvinar_l	-22	-28	12	83	8.334	nOCD < rOCD, rOCD > HC
SomMotA_4_l	Ventral Posterolateral Nucleus_r	16	-18	04	129	8.511	nOCD < rOCD, rOCD > HC
	Caudate_r	20	-02	18	87	9.276	nOCD < rOCD, rOCD > HC
Thalamus_r	Cerebellar Crus I_r	40	-68	-38	264	9.144	nOCD > rOCD, nOCD > HC
	Postcentral Gyrus_l	-44	-28	68	116	8.486	nOCD < rOCD
SomMotB_4_r	Middle Temporal Gyrus_r	46	-70	-02	140	7.582	nOCD < rOCD
	Middle Occipital Gyrus_l	-38	-74	00	108	9.476	nOCD < rOCD, rOCD > HC

MNI Peak, Coordinates of primary peak locations in the Montreal Neurological Institute space; l, left; r, right; SomMot, somatomotor network; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control.

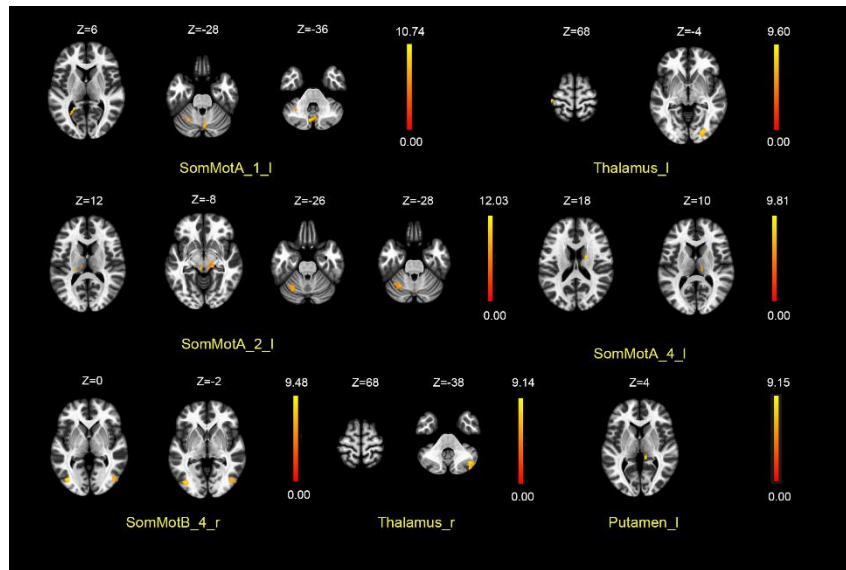


Figure 1. Brain regions with significant FC differences among the nOCD, rOCD and HC groups based on ANOVA analysis. The color bars indicate the F -value. SomMot, somatomotor network; l, left; r, right.

Post hoc analyses

We extracted differential FC values among the three groups and conducted post hoc analyses with Bonferroni correction ($p < 0.05$) for pairwise comparisons. Results are summarized in Table 2 (complete results in Supplementary Table 5).

Across all analyses, we found a consistent pattern of altered FC within the sensorimotor circuit (Figure 2A). Compared to HC, the rOCD group showed significantly increased FC including Thal_l-PoG_l, SomMotA_2_l-STN, SomMotA_4_l-VPL_r and SomMotA_4_l-Caud_r. Although the difference did not reach statistical significance, the connectivity between Thal_r and PoG_l also showed an increasing trend in the rOCD group relative to HC. In contrast, all the aforementioned FC values were decreased in the nOCD group compared to the HC, though not statistically significant.

Another finding was the presence of several altered FC between the sensorimotor circuit and cerebellum (Figure 2B). The nOCD group showed a significantly increased connectivity between Thal_r and Crus1_r, and significantly decreased FC values including SomMotA_1_l-VI_l and SomMotA_2_l-VI_l, relative to the HC group. In the rOCD group, the connectivity between SomMotA_1_l and VIIb_l was significantly increased compared to HC group.

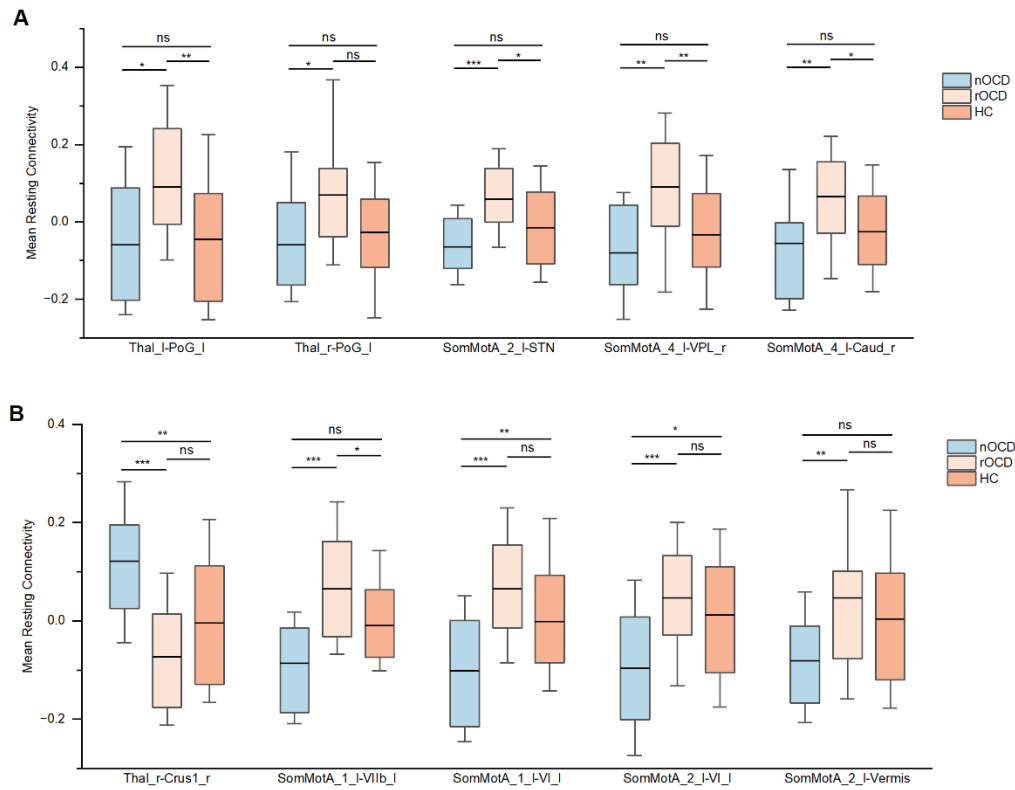


Figure 2. Bar plots for partial results of post hoc analyses of FC values between each pair group (nOCD group vs. rOCD group, nOCD group vs. HC, rOCD group vs. HC). The y-axis shows the values extracted from the significant FC, the central line represents the mean value, the box spans from the 25th to the 75th percentile, and the whiskers extend from the 10th to the 90th percentile. (A) Differences in FC within the sensorimotor circuit. (B) Differences in FC between the sensorimotor circuit and cerebellum. Significance coding: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns $p > 0.05$. Thal, thalamus; PoG, postcentral gyrus; SomMot, somatomotor network; STN, subthalamic nucleus; VPL, ventral posterolateral nucleus; Caud, caudate; Crus1, cerebellar Crus I; VIIb, cerebellar lobule VIIb; VI, cerebellar lobule VI; Vermis, cerebellar vermis VII; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control; l, left; r, right.

Correlation analysis between functional connectivity and clinical features

To explore the relationship between altered FC and clinical features in OCD patients, we conducted Pearson correlation analyses both across and within the nOCD and rOCD groups. No significant correlations were found between baseline Y-BOCS scores and any altered FC either across the two groups or within the rOCD group. However, in the nOCD group, baseline Y-BOCS scores were positively correlated with FC values between SomMotA_1_l and VI_l (Supplementary Figure 1). Several correlations were identified between Δ Y-BOCS scores and differential FC

(Supplementary Figure 2), with the majority of associations being positive.

Predicting treatment response to SSRIs

Significant predictive variables associated with treatment response are presented in Figure 3 and Supplementary Table 6. After FDR correction, no clinical variables showed significant predictive value. In contrast, all altered FC between the nOCD and rOCD groups exhibited statistically significant and potentially meaningful predictive capacities, with OR ranging from less than 0.001 to over 70,000 and R^2 ranging from 0.144 to 0.479 (Figure 3A).

Among these, only three FC values—Thal_r-Crus1_r, Put_l-WM and SomMotA_1_l-CC—were positively associated with poor response (Figure 3B), with Thal_r-Crus1_r showing the strongest predictive effect (OR < 0.001, R^2 = 0.479, AUC = 0.860). All remaining FC values were positively associated with treatment response (Figure 3C), among which SomMotA_1_l-VIIb_l (OR = 70228.218, R^2 = 0.374, AUC = 0.820) and SomMotA_1_l-VI_l (OR = 69519.668, R^2 = 0.420, AUC = 0.823) exhibited the most notable predictive strength.

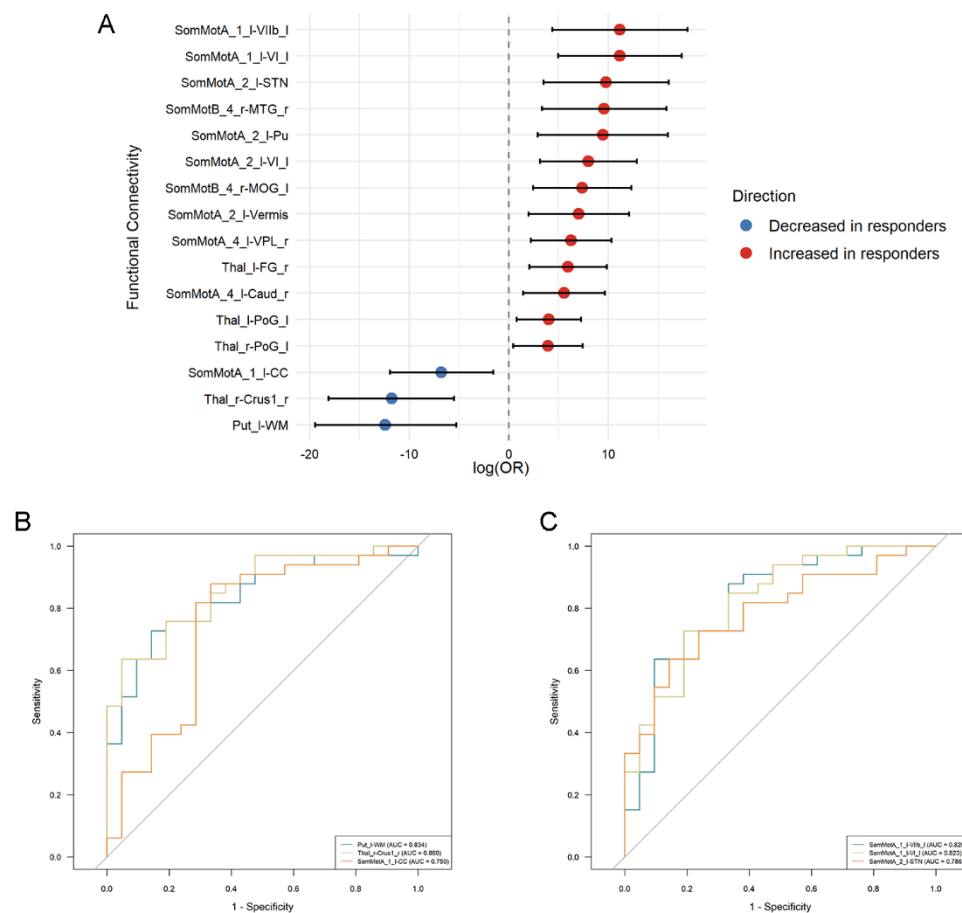


Figure 3. Predictive values of FC for SSRI treatment response in OCD patients. (A) Forest plot

showing ORs and 95% CI for altered FC between rOCD and nOCD groups identified by binary logistic regression analyses. FC associated with treatment response is shown in red, and those associated with non-response are shown in blue. (B) Receiver operating characteristic (ROC) curves for FC associated with non-response. (C) ROC curves for representative FC with high predictive performance associated with treatment response. OR, the odds ratio; AUC, area under the curve; Thal, thalamus; CrusI, cerebellar Crus I; PoG, postcentral gyrus; Put, putamen; WM, white matter; FG, fusiform gyrus; SomMot, somatomotor network; VIIb, cerebellar lobule VIIb; VI, cerebellar lobule VI; CC, corpus callosum; STN, subthalamic nucleus; Vermis, cerebellar vermis VII; Pu, pulvinar; VPL, ventral posterolateral nucleus; Caud, caudate; MTG, Middle Temporal Gyrus; MOG, Middle Occipital Gyrus; l, left; r, right.

Discussion

Reliable and clinically applicable predictors for SSRI response will serve to reduce disability and socioeconomic burden of OCD patients. The sensorimotor circuit, implicated in OCD pathology and targeted by serotonergic medication, may have a potential predictive value. Using seed-based rs-fMRI targeting the sensorimotor circuit, we found distinct FC patterns across rOCD, nOCD, and HC groups. Altered FC mainly involved connections within the sensorimotor circuit, as well as between the circuit and the cerebellum. Moreover, these FC values could discriminate patients by SSRI response (responders vs non-responders), suggesting their potential as neural biomarkers for SSRI response in OCD.

Distinct FC patterns within the sensorimotor circuit as a biomarker for SSRI response

Within the sensorimotor circuit, FC that differed between rOCD and nOCD groups displayed a consistently altered direction, for most FC values in the rOCD group were significantly increased compared to HC, while the nOCD group exhibited a trend in the opposite direction, though not statistically significant. Specifically, altered connectivity was observed between the bilateral thalamus and left PoG, as well as between the left sensorimotor cortices and subcortical structures in this circuit, such as STN, VPL and caudate, which play roles in the sensorimotor process[39-41]. FC values were significantly increased in the rOCD group compared to HC, while the nOCD group exhibited a trend in the opposite direction, suggesting this state of sensorimotor hyper-coupling in the rOCD group may contribute to excessive sensorimotor integration and cause impetuous, repetitive motor routine—features of compulsive. In contrast, the reduced integration of the circuit in the nOCD group may indicate a lower capacity for neural function. The differential patterns of FC observed among responders and non-responders may also partially explain the heterogeneity in

previous studies investigating FC within the sensorimotor circuit in OCD patients. Several rs-fMRI studies, including the research from the ENIGMA consortium, have observed decreased FC within the sensorimotor regions[13,42-44]. Conversely, increased FC values within sensorimotor circuits[45,46], such as elevated FC values between the left STN and right premotor cortex[47], has also been reported. Such discrepancies may arise from methodological differences, medication status, or clinical heterogeneity. Our findings highlighted that inconsistencies may also reflect intrinsic heterogeneity in neural coupling and functional preservation, associated with SSRI treatment response.

Studies in populations other than OCD patients have also examined the relationship between SSRIs and the sensorimotor circuit. Among healthy volunteers, sertraline administration reduced connectivity between the sensorimotor cortices and regions such as the anterior cingulate cortex and SMA[48]. Similarly, patients with depression showed decreased resting-state activity in sensorimotor cortices following SSRI treatment[49]. These findings, along with evidence that SSRIs modulate neuroplasticity within large-scale brain networks, including the sensorimotor circuit[25,50], give way to a potential mechanism by which SSRIs may exert their therapeutic effects. Specifically, the increased connectivity observed in rOCD group may indicate better preservation of functional capacity and neural integrity. Given the dense serotonergic innervation in this circuit and its role in sensorimotor integration[20,21], such preservation could facilitate engagement with SSRI-induced neuroplastic processes. This enhanced responsiveness may in turn support recovery through gradual recalibration of sensory–motor coordination during daily behavior. This proposed mechanism remains hypothetical and validation through longitudinal and multimodal investigations will be essential to clarify the neurobiological basis of treatment heterogeneity in OCD.

Cerebellar coupling with the sensorimotor circuit

Another notable and unexpected finding in our study was the significant differential FC between the cerebellum and the sensorimotor circuit, highlighting a potentially underappreciated role of the cerebellum in OCD. Altered connectivity was observed between the right thalamus and cerebellum Crus I, as well as between left sensorimotor cortices and cerebellar lobule VIIb, lobule VI and vermis VII. Traditionally, the cerebellum is considered playing a critical role in the sensorimotor system[51,52], but accumulating evidence suggests that the cerebellum also contributes to higher-level brain functions, including cognition, emotion regulation, and language processing[53,54]. Consequently, it has emerged as a novel and

increasingly important target of investigation in OCD research. Prior studies found that cerebellar lesions can trigger the onset of OCD symptoms[55]. In addition, neuroimaging studies have revealed cerebellar structural and functional abnormalities in OCD patients, including increased gray matter volume[56], reduced fractional anisotropy[57], and altered regional spontaneous brain activity[58]. However, current research investigating the relationship between SSRI treatment efficacy and cerebellar-cerebral networks remains limited. Our findings suggest that connectivity between the cerebellum and the sensorimotor circuit may play a role in the neural regulation underlying treatment response in OCD. Given the capacity for forward modulation of cerebellum[59,60], we hypothesize that its integrative and regulatory influence over sensorimotor and other brain networks may be important for SSRI efficacy, while it warrants further empirical validation.

Predictive utility and clinical translation

Identifying reliable predictors of SSRI treatment response in OCD remains a major clinical challenge. In our study, all differential FC between the rOCD and nOCD groups—rather than clinical characteristics—were predictive of treatment outcomes, with notably high ORs and satisfactory R^2 values. Unlike previous studies that primarily focused on FC features involving the raphe nucleus and cerebellum[26,27], we selected the sensorimotor circuit as the ROI due to its closer alignment with clinical phenomena[12,61], tight integration within the CSTC loop, and its extensive targeting in neuromodulation therapies for OCD. However, despite the strong predictive effects suggested by the large ORs, caution is warranted given the relatively small sample size and the wide 95% CI. Therefore, external validation in independent datasets is essential to confirm the generalizability of these results. Taken together, our findings suggest that the FC of sensorimotor circuit hold promise as predictive biomarker for SSRI treatment response in OCD, with potential utility in guiding personalized treatment strategies.

Limitation

Several limitations should be noted. First, although we found that the patients with different treatment responses to SSRIs had abnormal FCs within the sensorimotor circuit, we were unable to directly relate them to clinical manifestations. This was due to the absence of specific clinical or cognitive assessments targeting sensorimotor-related processes, such as the presence of SP or impairments in sensorimotor gating. Therefore, further studies should consider integrating neuroimaging data with clinical features to better elucidate these associations. Second,

our hypothesis that SSRIs may exert their pharmacological effects, at least in part, through modulation of the sensorimotor circuit could not be directly tested, as no post-treatment neuroimaging data were available. This warrants further investigation with longitudinal imaging designs. Third, although the abnormal FCs observed in our study achieved promising classification performance, the absence of an external validation dataset limits the generalizability and clinical applicability of them. Future studies should incorporate external datasets to validate the robustness and clinical utility of the predictors.

Conclusion

In conclusion, we identified that baseline FC in the sensorimotor circuit could effectively discriminate SSRI responders from non-responders among drug-naïve OCD patients. The observed hyperconnectivity in responders suggested a preserved neural substrate that might facilitate SSRI-induced neuroplasticity, whereas hypoconnectivity in non-responders might reflect impaired circuit integrity less amenable to serotonergic modulation. These findings position the sensorimotor circuit not only as a core component of OCD pathophysiology but also as a promising predictor of pharmacological treatment outcome. Future studies should validate these biomarkers in independent cohorts and explore their utility in guiding targeted interventions, ultimately advancing toward personalized treatment for OCD.

CRedit Authorship Contribution Statement

Tingyue Bai: Writing – original draft, Writing – review & editing, Formal analysis, Investigation, Data curation; **Hanyang Ruan:** Writing – review & editing, Conceptualization, Methodology, Validation; **Yang Wang:** Writing – review & editing, Conceptualization, Methodology; **Gui Zhang:** Conceptualization; **Jian Gao:** Investigation; **Zifeng Zheng:** Investigation; **Di Li:** Investigation; **Qing Zhao:** Investigation; **Huiqin Han:** Resources; **Qing Fan:** Resources; **Dongdong Shi:** Resources; **Zhen Wang:** Supervision, Resources, Funding acquisition, Conceptualization.

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Data availability statement

Individual-level clinical and imaging data derive from an ongoing longitudinal study, and therefore they cannot be publicly shared until the cohort's primary results are published. Derived measures and analysis code are available from the corresponding author upon reasonable request.

Competing Interests

The authors declare no competing interests.

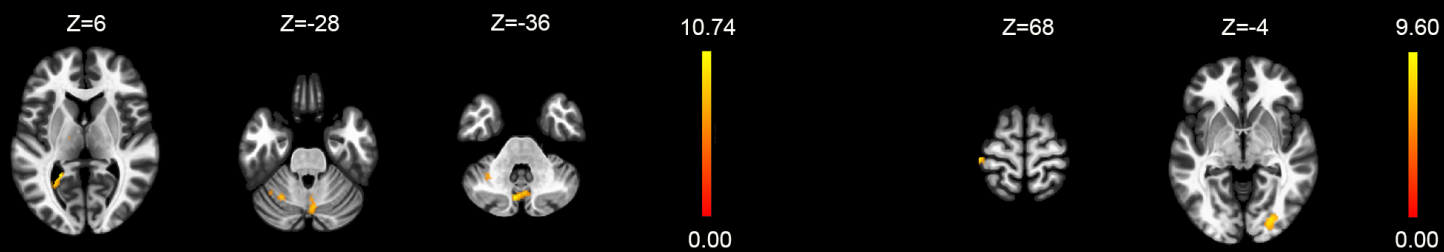
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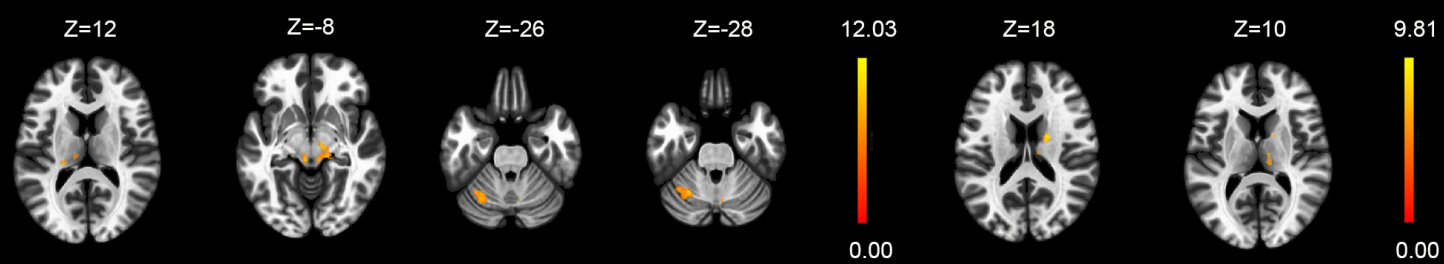
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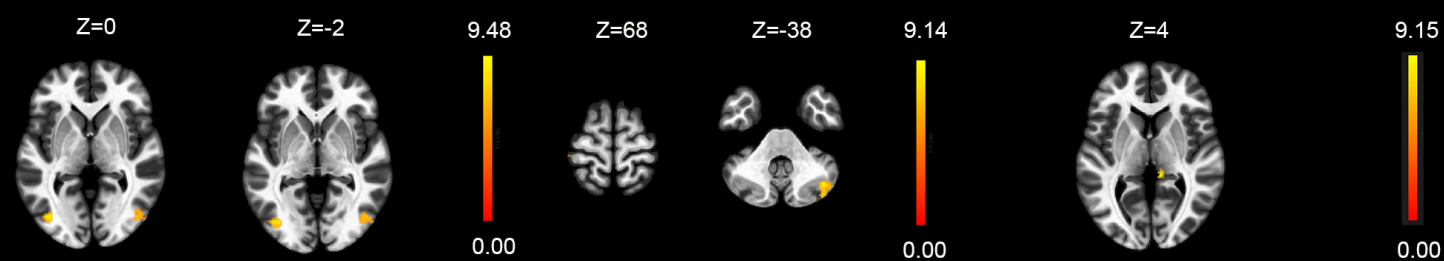
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Thalamus_l



SomMotA_2_l

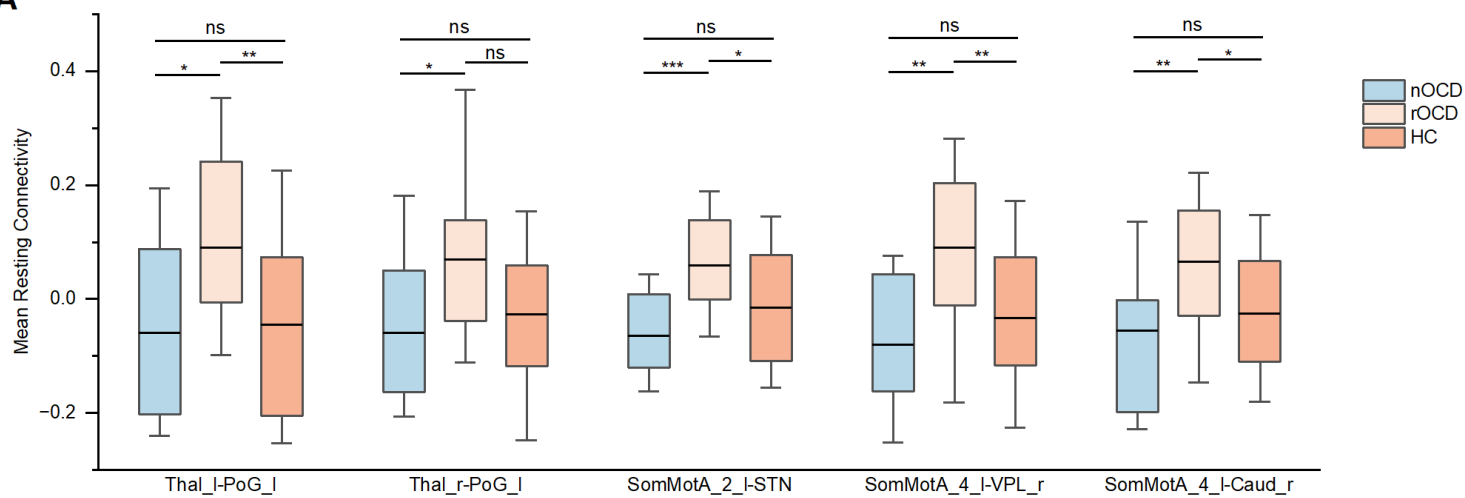
SomMotA_4_l



SomMotB_4_r

Thalamus_r

Putamen_l

A**B**